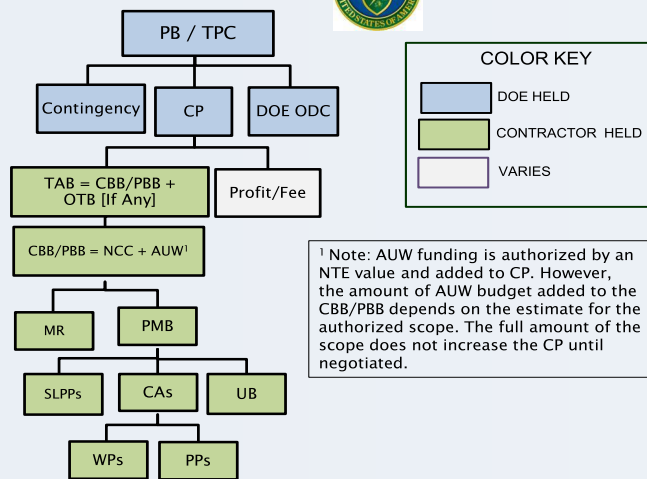
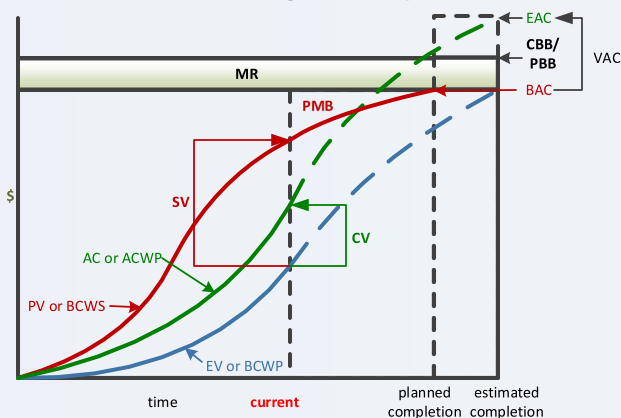




Earned Value Management System Basics



Performance Baseline Components

(Performance Baseline must clearly document scope and CD-4 date)

AUW = Authorized Unpriced Work (contractually approved, but not yet negotiated)
CA = Control Account (includes AUW) = WPs + PPs
CBB = Contract Budget Base = PMB + MR; valid when 1 contract to 1 project; else PBB
CP = Contract Price = CBB + Profit/Fee
MR = Management Reserve is held by contractor (Contingency is held by DOE)
NCC = Contract price less Profit/Fees
ODC = Other Direct Costs
OTB = Established performance budget that exceeds the value of the negotiated contract
PB = Performance Baseline (TPC) = CP + Contingency + DOE ODC
PBB = Project Budget Base = PMB + MR; valid when 1 contract to multiple projects
PMB = Performance Measurement Baseline = CAs + UB + SLPPs
PP = Planning Package (far-term activities within a CA)
SLPP = Summary Level Planning Package
TAB = Total Allocated Budget CBB + OTB or PMB + MR + OTB
TPC = Total Project Cost
UB = Undistributed Budget (activities not yet distributed to CA)
WP = Work Package (near-term, detail-planned activities within a CA)

EVMS Basic Components

AC = Actual Cost = ACWP = Actual Cost of Work Performed
EV = Earned Value = BCWP = Budgeted Cost for Work Performed
PV = Planned Value = BCWS = Budgeted Cost for Work Scheduled
BAC = Budget at Completion = $\sum BCWS$ = Sum of Budgeted Cost for Work Scheduled
EAC = Estimate at Completion = ACWP + Estimate to Complete (ETC)

VARIANCES

CV = EV - AC = BCWP - ACWP = Cost Variance
SV = EV - PV = BCWP - BCWS = Schedule Variance
CV% = (EV - AC) / EV = (BCWP - ACWP) / BCWP = Cost Variance (%)
SV% = (EV - PV) / PV = (BCWP - BCWS) / BCWS = Schedule Variance (%)
VAC = BAC - EAC = Variance at Completion

OVERALL STATUS

% scheduled = PV_{cum} / BAC = BCWS_{cum} / BAC
% complete = EV_{cum} / BAC = BCWP_{cum} / BAC
% budget spent = AC_{cum} / BAC = ACWP_{cum} / BAC
Work Remaining (WR) = BAC - EV_{cum} = BAC - BCWP_{cum}

PERFORMANCE INDICES (Favorable is >1.0, unfavorable is <1.0)

CPI = EV / AC = BCWP / ACWP = Cost Performance Index
SPI = EV / PV = BCWP / BCWS = Schedule Performance Index
TCPI_{EAC} = WR / (EAC - AC_{cum}) = EAC-based To Complete Performance Index

ESTIMATE AT COMPLETION FORMULAE

EAC = BAC / CPI_{cum} = Estimate at Completion (general)
EAC_{CPIcum} = AC_{cum} + WR / CPI_{cum} = Estimate at Completion (CPI)
EAC_{composite} = AC_{cum} + WR / (CPI_{cum} * SPI_{cum}) = Estimate at Completion (composite)
EAC_{CPI3mo} = AC_{cum} + WR / CPI_{3mo} = Estimate at Completion (3 Mo. CPI)
Note: CPI_{3mo} = (IncEVn + IncEVn-1 + IncEVn-2) / (IncACn + IncACn-1 + IncACn-2)